Review

Monday, 17 February 2025 12:35 PM

Linear Algebra

$$A\vec{x} = \vec{b}$$

- Unique solution exists if A is non-singular (i.e. if $\det(A) \neq 0$)

Eigenvalues of N×N matrix : $\det(A - \lambda I) = 0$; $A x_i = \lambda_i x_i$ (x_i: eigenvector)

- There are N eigenvalues ; depending on A they might be real or complex (i.e. rank N will have N real eigenvalues & all eigenvectors are orthogonal)
  - symmetric → $a_{ij} = a_{ji}$ & square

Eigenvalues tell us how a system behaves, evolves and transforms
↓
i.e. They provide deep insights into the matrix's properties and behavior
eg: → how a linear transformation will stretch or compress vectors
→ dynamical stability
→ observable quantities such as energy levels etc.

- Positive definite : A is positive definite if $\vec{x}^T A \vec{x} > 0$, $x \neq 0$
  ↳ All eigenvalues are positive

- Spectral radius : of a square matrix $\rho(A)$ is the largest absolute value of its eigenvalues
  ↳ $\rho(A) \leq \|A\|$

Methods of solving $Ax = b$

Direct methods

- Diagonal matrix
- Triangular "
  → forward / backward sub
- Cramer's rule $O(N!)$
- Gaussian elimination ($\frac{N^3}{3} + O(N^2)$ arithmetic operations)
  → Gauss - Jordan ($\frac{N^3}{2} + O(N^2)$)
  → LU decomposition
  → Thomas algorithm (for tridiagonal matrix)

banded = diagonally dominant.

require pivoting & equilibration (unless banded or positive definite)

Iterative Methods

- Jacobi iteration [$A = D + L' + U'$]
- Gauss - Seidel iteration [$A = D(L'' + I + U'')$]

convergence guaranteed if A is banded or positive definite

Error / Accuracy

All computations have round off errors → hence you get approximate solutions
↓
need to assess accuracy by comparing magnitude/size of different solutions
↓
Norms (works for scalars, vectors, tensors)

$$L_p(x) = \|x\|_p = \left(\sum_{i=1}^{N} |x_i|^p\right)^{1/p} \quad \text{(unweighted)}$$

$$L_p(x) = \|x\|_{p,w} = \left(\sum_{i=1}^{N} w_i |x_i|^p\right)^{1/p} \quad \text{(weighted)}$$

a = approx.
e = exact

$A_a = A_e + \delta A$

$x_a = x_e + \delta x$

$A_a x_a = b_e$

$$\text{Relative error} = \frac{\|\delta x\|}{\|x_e + \delta x\|} \leq \kappa(A) \frac{\|\delta A\|}{\|A_e\|}$$

$\kappa(A)$: Spectral condition number $\left(= \frac{\lambda_{max}}{\lambda_{min}}\right)$
(measure of how sensitive the solution of a linear system $Ax = b$ is to

Spectral condition number $\left(= \frac{\lambda_{max}}{\lambda_{min}}\right)$

(measure of how sensitive the solution of a linear system $Ax=b$ is to small changes in b)

$\kappa(A) \approx 1$ (well-conditioned)

If $\kappa(A) >> 1$ (ill-conditioned)

↳ Iterative refinement: improving accuracy of the solution

- Calculate residual (using twice precision) $\vec{r}$
- Calculate error making this residual LU $\vec{\delta x} = \vec{r}$
- Subtract error from original solution $x_a = x_a - \delta x$
- Check $\left|\left| \frac{\delta x}{x_a} \right|\right| < \varepsilon$ [relative error]
- End loop

Partial Differential Equations

Phenomena where changes occur over multiple variables → very real-world applicable

$$F(x, y, \ldots, U_x, U_y, \ldots, U_{xx}, U_{yy}, U_{xy} \ldots) = 0$$

U_x ↳ $\frac{\partial U}{\partial x}$; U_{xx} ↳ $\frac{\partial U}{\partial x^2}$

↓ functional relationship among independent vars, dependent var & the partial derivatives

U → dependent var

x, y → independent vars

- Order → determined by highest-order PDE in the entire expression

1) Linear → F is a linear function of the dependent variable & its partial derivatives

$$x U_x + x_y U_y + \underbrace{U_{xy}}_{2^{nd}\ order} = 0$$

2) Almost linear → small or mild non-linearities

3) Quasi-linear → F is linear in highest order partial derivative, but non-linear in the dependent variable and/or lower-order derivatives

$$\underbrace{U U_x}_{\text{non-linearity}} + U_y = x_y$$

but U_x & U_y are linear (i.e. no U_x^2 or $\sin(U_x)$)

4) Non-linear → F is non-linear in any or all of the highest order partial derivatives appearing in the equation

$$U U_x + U_y^2 = x_y$$

E.g: ↳ $U^2 U_{xx} + U_y = x y^2$ → quasi-linear

↳ $U_x U_{xy} = \sin(xy)$ → Nonlinear

↳ This doesn't matter since U & its derivatives are what we check for

Solving:

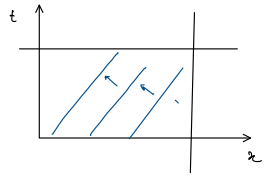
- First-order linear + quasi-linear

↳ Characteristic curves (by converting system of ODE's where family of curves in the $x_i - U$ space)

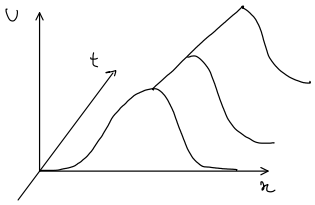
↓

along the curves, PDE is converted to ODE requiring one BC / IC

eg: $U_t + cU_x = 0 \rightarrow x = c_o + ct$







Weak vs. strong solutions

System of ODE's: $U_t + AU_x = 0$ (A: eigenvalues)
- → Elliptic
- → Hyperbolic
- → Mixed

• Second-order linear & quasi-linear PDE's

real or complex characteristic curves; may have two curves passing through each point

$$a_{ij} U_{x_i x_j} + b = 0$$

a_{ij} & b are functions only of x_i, U and U_x → quasi-linear

↳ Quasi-linear are further defined: $A = a_{ij}$

problems w/ evolutionary character:
- → Ultrahyperbolic (all eigenvalues are non-zero & at least two positive & two negative)
- → Hyperbolic (all eigenvalues are non-zero & all but one has same sign)
- → Parabolic (at least one eigenvalue is zero)

equilibrium problems:
- → Elliptic (All eigenvalues of A have same sign)
- → Mixed